

Nathan Ko

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EDUCATION

University of Waterloo

Bachelor of Mathematics, Financial Analysis & Risk Management (FARM)

Waterloo, ON

Sept 2025 – Present

- Intended Minor: Statistics
- Relevant Coursework: Multivariable Calculus, Linear Algebra, Functional Programming, Algorithms & Data Abstraction, Probability, Financial Mathematics, Financial Accounting

EXPERIENCE

Technology Analyst

Government of Ontario

May 2026 – Aug 2026

Toronto, ON

- Designed and implemented **AI-driven automation workflows** and a reusable prompt library to **streamline software testing processes**, improving efficiency and consistency.
- Analyzed business requirements and executed **100+ functional and regression tests**, identifying implementation issues and supporting solution delivery in an Agile environment.
- **Bridged software & business teams** to clarify requirements, resolve implementation issues, and deliver reliable enterprise software.

Software Engineer

CYC Motor

Jul 2025 – Aug 2025

Hong Kong

- Built and deployed an **AI-powered** customer support chatbot using a **retrieval-augmented generation (RAG)** pipeline over internal product manuals, FAQs, and support documentation.
- Integrated the chatbot into the company website, serving **2,000+ monthly user interactions**, deflecting a significant portion of human support tickets.
- Engineered the chatbot's retrieval pipeline and prompt architecture to **reduce hallucinations** while delivering context-aware answers across the company's product catalog.

Co-Founder

Aim High Tutoring

Sept 2024 – June 2025

West Vancouver, BC

- Founded and scaled a peer-to-peer math tutoring business from **0 to 18 clients** in 9 months, generating **\$15K+ CAD in revenue** with zero external funding.
- Developed a client acquisition platform using **Javascript/PostgreSQL (Supabase)** for lead capture and testimonial management, improving conversion rates by **150%**.
- Managed operations for a team of **4 tutors**, overseeing hiring, scheduling, client onboarding, and finances.

PROJECTS & EXTRACURRICULAR

Tech Layoff Stock Impact Predictor | *Python, Pandas, Streamlit, scikit-learn, XGBoost*

- Built an end-to-end **ML pipeline** predicting stock drawdowns from tech layoff events using **4,000+ records** merged with Yahoo Finance data via Pandas.
- Deployed an interactive Streamlit dashboard visualizing model predictions, **SHAP feature importance**, and performance metrics (**AUC-ROC, precision/recall**).

Varsity Team Member | *Waterloo Men's Varsity Squash*

- Held the **#1 spot** on the varsity roster, committing **20+ hours weekly** alongside a full academic course load.
- Led team to **4th place finish** at Ontario University Athletics Championships; afterwards individually placing **13th in U19 national championship**.

SKILLS

Programming Languages: Python, SQL, TypeScript, C, HTML/CSS

Frameworks & Tools: Pandas, NumPy, scikit-learn, PostgreSQL, React, Git, Azure DevOps, Power BI, Excel

Spoken Languages: English, Cantonese, Mandarin

Hobbies/Interests: Golf, Running, Weightlifting, Cliff Jumping, Chess, Poker, Sudoku, Philosophy